Akshat Lal

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Education

University of Waterloo Sept 2026 – May 2031
Bachelor of Applied Science, Computer Engineering Waterloo, ON

Engineering Experience

VEX Robotics Team 2022 – 2026
Team Lead

- Competed **4 consecutive seasons**, progressing from team member to **Team Lead** and owning the full **design, build, test, and iteration** cycle for each competition robot
- Set **mechanical design, assembly, and programming** priorities across subteams, coordinating integration of mechanisms and control code into a working competition robot each season
- Ran **pre-competition reliability testing** to locate failure points and fix them before events, **reducing in-match breakdowns**

Student Design Team Sept 2026 – Present
Member, Electrical Team

- Designing and laying out **printed circuit boards** in **Altium Designer** for **onboard vehicle electronics**; preparing to contribute to the electrical design of a **moving mechanical arm subsystem**
- Participating in **design reviews of component selection and PCB layout** with senior electrical and mechanical members, learning the team's design workflow ahead of the arm subsystem build

Projects

Caissa — Chess Engine and Desktop Application | *C#, .NET, WinForms*

- Architected a **C# chess engine** and **WinForms desktop app** from scratch using **object-oriented** board, piece, move, and game-state models that are reused across the engine and UI
- Implemented **move generation and legality checking** (movement, check, checkmate, captures) for **all 6 piece types** across **3 rule sets** (Standard, Chess960, Atomic), so **one engine drives every variant**
- Built an **event-driven UI layer** with **move highlighting, game clocks, and move history** that renders live game state to the player
- Developed an **AI opponent** by writing a **move-evaluation function** and integrating it with the game-state and UI layers, enabling full games against the computer

Personal Portfolio Website | *Next.js, React, TypeScript, CSS, Vercel*

- Built a **responsive** portfolio site with **Next.js, React, and TypeScript** to showcase projects and skills
- Implemented a **draggable desk-style interface** with custom navigation and a **light/dark theme toggle** in React and CSS
- Set up a **CI/CD pipeline** on **Vercel** so every pushed commit is automatically built and deployed

Task Management Web App | *TypeScript, React, Next.js, SQL*

- Developed a **full-stack task manager** in **Next.js and TypeScript** supporting task creation, editing, organization, and status tracking
- Designed a **relational SQL schema and persistence layer** so tasks and project state persist across sessions

Java Image Editor | *Java, Swing, OOP*

- Built a **desktop image editor** in **Java Swing**, separating image data, editing operations, and UI into distinct classes so new features are easy to add
- Wired **event-driven GUI controls** to the editing operations so edits apply interactively and update the canvas immediately
- Implemented **pixel-level image algorithms** for **grayscale, cropping, rotation, and brightness adjustment**

Technical Skills

Languages: C#, Java, C++, Python, TypeScript, JavaScript, HTML, CSS, SQL

Frameworks & Tools: .NET, WinForms, React, Next.js, Arduino, Git, Vercel, Altium Designer

Concepts: OOP, Data Structures & Algorithms, API Integration, GUI Development, Debugging, PCB Design